

Lecture 8

Mathematical problems

Example 1: A particle performs simple harmonic motion given by the equation.

$$y = 20 \sin (\omega t + \alpha)$$

If the time period is 30 seconds and the particle has a displacement of 10 cm at $t=0$, find (i) epoch, (ii) the phase angle at $t= 5$ seconds and (iii) the phase difference between two position of the particle 15 seconds apart.

Solution:

Here

$$y = 20 \sin (\omega t + \alpha)$$

$$T=30 \text{ seconds}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{30} = \frac{\pi}{15} \text{ rad/sec}$$

(i) at $t=0$, $y= 10$ cm

$$10 = 20 \sin \left(\frac{\pi}{15} \times 0 + \alpha \right)$$

$$\text{or, } \sin \alpha = \frac{10}{20} = 0.5$$

$$\text{or, } \alpha = \sin^{-1}(0.5)$$

$$\text{or, } \alpha = \frac{\pi}{6} \text{ radian.}$$

(ii) at $t= 5$ sec

The phase angle $= (\omega t + \alpha)$

$$= \left(\frac{\pi}{15} \times 5 + \frac{\pi}{6} \right) = \frac{\pi}{2}$$

(iii) at $t=0$ the phase angle

$$\theta_1 = \left(\frac{\pi}{15} \times 0 + \frac{\pi}{6} \right) = \frac{\pi}{6} \text{ radian}$$

at $t = 15$ sec, the phase angle

$$\theta_2 = \left(\frac{\pi}{15} \times 15 + \frac{\pi}{6} \right) = \frac{7\pi}{6} \text{ radian}$$

$\therefore$ the phase difference,

$$\theta_2 - \theta_1 = \frac{7\pi}{6} - \frac{\pi}{6} = \pi \text{ radian.}$$

Example 2: A simple harmonic wave of amplitude 8 units traverse a line of particle in the direction of positive x-axis. At any given instant of time, the displacement of a particle at a distance of 10 cm from the origin +6 units while that for a particle at a distance of 25 cm from the origin + 4 units. Calculate the wavelength.

Solution:

The equation of a simple harmonic wave can be written as

$$y = a \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\text{Or, } y = a \sin 2\pi \left(\frac{vt}{\lambda} - \frac{x}{\lambda} \right) [v = n\lambda]$$

$$\text{Or, } y = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right)$$

$$\text{or, } y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) [n = \frac{1}{T}]$$

$$\text{Or, } \frac{y}{a} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

In the first case,

$y = +6$, $a = 8$ and $x = 10$ cm

$$\therefore \frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) \dots\dots\dots (1)$$

In the second case,

$y = +4$, $a = 8$ and $x = 25$ cm

$$\therefore \frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right) \dots\dots\dots (2)$$

From equation (1)

$$\frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right)$$

$$\text{or, } 0.75 = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right)$$

$$\text{or, } \sin^{-1}(0.75) = 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right)$$

$$\text{or, } 48.6 = 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right)$$

$$\text{or, } \left(\frac{t}{T} - \frac{10}{\lambda} \right) = \frac{48.6}{2\pi}$$

$$\text{or, } \left(\frac{t}{T} - \frac{10}{\lambda} \right) = \frac{48.6}{360} \dots\dots\dots (3)$$

From equation (2)

$$\frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right)$$

$$\text{or, } 0.5 = \sin 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right)$$

$$\text{or, } \sin^{-1}(0.5) = 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right)$$

$$\text{or, } 30 = 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right)$$

$$\text{or, } \left(\frac{t}{T} - \frac{25}{\lambda} \right) = \frac{30}{2\pi}$$

$$\text{or, } \left(\frac{t}{T} - \frac{25}{\lambda} \right) = \frac{30}{360} \dots\dots\dots (4)$$

Subtracting equation (4) from (3)

$$\left(\frac{t}{T} - \frac{10}{\lambda}\right) - \left(\frac{t}{T} - \frac{25}{\lambda}\right) = \frac{48.6}{360} - \frac{30}{360}$$

$$\text{or, } \frac{t}{T} - \frac{10}{\lambda} - \frac{t}{T} + \frac{25}{\lambda} = \frac{48.6-30}{360}$$

$$\text{or, } \frac{25}{\lambda} - \frac{10}{\lambda} = \frac{18.6}{360}$$

$$\text{or, } \frac{1}{\lambda} (25-10) = \frac{18.6}{360}$$

$$\text{or, } 15 = \lambda \times \frac{18.6}{360}$$

$$\text{or, } \lambda = \frac{15 \times 360}{18.6}$$

$$\text{or, } \lambda = 290.32 \text{ cm.}$$

Example 3: Deduce the frequency and quality factor for a circuit with $L = 2\text{mH}$, $C = 5 \mu\text{F}$ and $R = 0.2 \text{ ohm}$

Solution:

The circuit is equivalent to a damped (or underdamped) harmonic oscillator. The angular frequency of oscillation is given by $g = \sqrt{(\omega^2 - \lambda^2)}$ where ω is the natural angular frequency of the oscillator and λ is the damping constant. Hence the frequency of oscillation

$$n = \frac{g}{2\pi} = \frac{\sqrt{(\omega^2 - \lambda^2)}}{2\pi}$$

$$\text{or, } n = \frac{1}{2\pi} \sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

Here $L = 2\text{mH} = 0.002 \text{ henry}$

$C = 5 \mu\text{F} = 5 \times 10^{-6} \text{ farad}$

R= 0.2 ohm

$$\begin{aligned}\therefore n &= \frac{1}{2\pi} \sqrt{\left(\frac{1}{0.002 \times 5 \times 10^{-6}} - \frac{(0.2)^2}{4 \times (0.002)^2} \right)} \\ &= \frac{1}{2\pi} \sqrt{(10^8 - 2.5 \times 10^3)} \\ &= 1.592 \times 10^3 \text{ cps}\end{aligned}$$

Since the resistance is small, the quality factor $Q = \frac{L\omega}{R} = \frac{Lg}{R}$

Where $g = 2\pi n$

$$\begin{aligned}&= 2\pi \times 1.592 \times 10^3 \\ &= 10^4 \text{ rad/sec.}\end{aligned}$$

$\therefore$ Quality factor for the circuit,

$$\begin{aligned}Q &= \frac{0.002 \times 10^4}{0.2} \\ &= 10^2 \\ &= 100\end{aligned}$$